

A Miyaoka-Yau inequality for hyperplane arrangements in $\mathbb{CP}^d$

Martin de Borbon (joint with Dmitri Panov)

Living on the edge of the moduli space, Aarhus

18-22 May 2026

Outline

- Introduction: constant curvature metrics on Riemann surfaces with conical singularities

- Introduction: constant curvature metrics on Riemann surfaces with conical singularities
- Main result: polyhedral Kähler metrics on $\mathbb{CP}^d$

- Introduction: constant curvature metrics on Riemann surfaces with conical singularities
- Main result: polyhedral Kähler metrics on $\mathbb{CP}^d$
- Sketch proof: Kobayashi-Hitchin correspondence for parabolic bundles

- Introduction: constant curvature metrics on Riemann surfaces with conical singularities
- Main result: polyhedral Kähler metrics on $\mathbb{CP}^d$
- Sketch proof: Kobayashi-Hitchin correspondence for parabolic bundles
- Conjectures: Kähler-Einstein metrics and Miyaoka-Yau inequality for log pairs

Metrics on surfaces with conical singularities

Metrics on surfaces with conical singularities

Let Σ be a compact orientable 2D surface.

Metrics on surfaces with conical singularities

Let Σ be a compact orientable 2D surface. Let $p_1, \dots, p_n \in \Sigma$ and let $\alpha_1, \dots, \alpha_n > 0$.

Metrics on surfaces with conical singularities

Let Σ be a compact orientable 2D surface. Let $p_1, \dots, p_n \in \Sigma$ and let $\alpha_1, \dots, \alpha_n > 0$. Let g be a metric on $\Sigma \setminus \{p_1, \dots, p_n\}$ with constant curvature $K = 1, 0, -1$ that near p_i is locally isometric to

$$\begin{cases} dr^2 + \alpha_i^2 \sin^2(r) d\theta^2 \\ dr^2 + \alpha_i^2 r^2 d\theta^2 \\ dr^2 + \alpha_i^2 \sinh^2(r) d\theta^2 \end{cases}$$

Metrics on surfaces with conical singularities

Let Σ be a compact orientable 2D surface. Let $p_1, \dots, p_n \in \Sigma$ and let $\alpha_1, \dots, \alpha_n > 0$. Let g be a metric on $\Sigma \setminus \{p_1, \dots, p_n\}$ with constant curvature $K = 1, 0, -1$ that near p_i is locally isometric to

$$\begin{cases} dr^2 + \alpha_i^2 \sin^2(r) d\theta^2 \\ dr^2 + \alpha_i^2 r^2 d\theta^2 \\ dr^2 + \alpha_i^2 \sinh^2(r) d\theta^2 \end{cases}$$

Examples: double of a polygon;

Metrics on surfaces with conical singularities

Let Σ be a compact orientable 2D surface. Let $p_1, \dots, p_n \in \Sigma$ and let $\alpha_1, \dots, \alpha_n > 0$. Let g be a metric on $\Sigma \setminus \{p_1, \dots, p_n\}$ with constant curvature $K = 1, 0, -1$ that near p_i is locally isometric to

$$\begin{cases} dr^2 + \alpha_i^2 \sin^2(r) d\theta^2 \\ dr^2 + \alpha_i^2 r^2 d\theta^2 \\ dr^2 + \alpha_i^2 \sinh^2(r) d\theta^2 \end{cases}$$

Examples: double of a polygon; surface of a polyhedron.

Metrics on surfaces with conical singularities

Let Σ be a compact orientable 2D surface. Let $p_1, \dots, p_n \in \Sigma$ and let $\alpha_1, \dots, \alpha_n > 0$. Let g be a metric on $\Sigma \setminus \{p_1, \dots, p_n\}$ with constant curvature $K = 1, 0, -1$ that near p_i is locally isometric to

$$\begin{cases} dr^2 + \alpha_i^2 \sin^2(r) d\theta^2 \\ dr^2 + \alpha_i^2 r^2 d\theta^2 \\ dr^2 + \alpha_i^2 \sinh^2(r) d\theta^2 \end{cases}$$

Examples: double of a polygon; surface of a polyhedron.

Gauss-Bonnet

$$\frac{1}{2\pi} \int_{\Sigma} K dA = \chi(\Sigma) + \sum_{i=1}^n (\alpha_i - 1)$$

Metrics on surfaces with conical singularities

Let Σ be a compact orientable 2D surface. Let $p_1, \dots, p_n \in \Sigma$ and let $\alpha_1, \dots, \alpha_n > 0$. Let g be a metric on $\Sigma \setminus \{p_1, \dots, p_n\}$ with constant curvature $K = 1, 0, -1$ that near p_i is locally isometric to

$$\begin{cases} dr^2 + \alpha_i^2 \sin^2(r) d\theta^2 \\ dr^2 + \alpha_i^2 r^2 d\theta^2 \\ dr^2 + \alpha_i^2 \sinh^2(r) d\theta^2 \end{cases}$$

Examples: double of a polygon; surface of a polyhedron.

Gauss-Bonnet

$$\frac{1}{2\pi} \int_{\Sigma} K dA = \chi(\Sigma) + \sum_{i=1}^n (\alpha_i - 1)$$

Metric + orientation $\implies$ Riemann surface X

McOwen (1988), Troyanov (1991)

McOwen (1988), Troyanov (1991)

Let X be a compact Riemann surface.

McOwen (1988), Troyanov (1991)

Let X be a compact Riemann surface. Let $x_1, \dots, x_n \in X$ and let $\alpha = (\alpha_1, \dots, \alpha_n) \in (\mathbb{R}_{>0})^n$.

McOwen (1988), Troyanov (1991)

Let X be a compact Riemann surface. Let $x_1, \dots, x_n \in X$ and let $\alpha = (\alpha_1, \dots, \alpha_n) \in (\mathbb{R}_{>0})^n$. Define

$$\chi(X, \alpha) = \chi(X) + \sum_{i=1}^n (\alpha_i - 1)$$

McOwen (1988), Troyanov (1991)

Let X be a compact Riemann surface. Let $x_1, \dots, x_n \in X$ and let $\alpha = (\alpha_1, \dots, \alpha_n) \in (\mathbb{R}_{>0})^n$. Define

$$\chi(X, \alpha) = \chi(X) + \sum_{i=1}^n (\alpha_i - 1)$$

- If $\chi(X, \alpha) < 0$ then there exists a unique Kähler metric on X with $K_g = -1$ and cone angle $2\pi\alpha_i$ at x_i

McOwen (1988), Troyanov (1991)

Let X be a compact Riemann surface. Let $x_1, \dots, x_n \in X$ and let $\alpha = (\alpha_1, \dots, \alpha_n) \in (\mathbb{R}_{>0})^n$. Define

$$\chi(X, \alpha) = \chi(X) + \sum_{i=1}^n (\alpha_i - 1)$$

- If $\chi(X, \alpha) < 0$ then there exists a unique Kähler metric on X with $K_g = -1$ and cone angle $2\pi\alpha_i$ at x_i
- If $\chi(X, \alpha) = 0$ then there exists a unique up to scale Kähler metric on X with $K_g = 0$ and cone angle $2\pi\alpha_i$ at x_i

McOwen (1988), Troyanov (1991)

Let X be a compact Riemann surface. Let $x_1, \dots, x_n \in X$ and let $\alpha = (\alpha_1, \dots, \alpha_n) \in (\mathbb{R}_{>0})^n$. Define

$$\chi(X, \alpha) = \chi(X) + \sum_{i=1}^n (\alpha_i - 1)$$

- If $\chi(X, \alpha) < 0$ then there exists a unique Kähler metric on X with $K_g = -1$ and cone angle $2\pi\alpha_i$ at x_i
- If $\chi(X, \alpha) = 0$ then there exists a unique up to scale Kähler metric on X with $K_g = 0$ and cone angle $2\pi\alpha_i$ at x_i
- If $\chi(X, \alpha) > 0$.

McOwen (1988), Troyanov (1991)

Let X be a compact Riemann surface. Let $x_1, \dots, x_n \in X$ and let $\alpha = (\alpha_1, \dots, \alpha_n) \in (\mathbb{R}_{>0})^n$. Define

$$\chi(X, \alpha) = \chi(X) + \sum_{i=1}^n (\alpha_i - 1)$$

- If $\chi(X, \alpha) < 0$ then there exists a unique Kähler metric on X with $K_g = -1$ and cone angle $2\pi\alpha_i$ at x_i
- If $\chi(X, \alpha) = 0$ then there exists a unique up to scale Kähler metric on X with $K_g = 0$ and cone angle $2\pi\alpha_i$ at x_i
- If $\chi(X, \alpha) > 0$. Furthermore, assume that $\alpha_i \in (0, 1)$ and $n \geq 3$.

McOwen (1988), Troyanov (1991)

Let X be a compact Riemann surface. Let $x_1, \dots, x_n \in X$ and let $\alpha = (\alpha_1, \dots, \alpha_n) \in (\mathbb{R}_{>0})^n$. Define

$$\chi(X, \alpha) = \chi(X) + \sum_{i=1}^n (\alpha_i - 1)$$

- If $\chi(X, \alpha) < 0$ then there exists a unique Kähler metric on X with $K_g = -1$ and cone angle $2\pi\alpha_i$ at x_i
- If $\chi(X, \alpha) = 0$ then there exists a unique up to scale Kähler metric on X with $K_g = 0$ and cone angle $2\pi\alpha_i$ at x_i
- If $\chi(X, \alpha) > 0$. Furthermore, assume that $\alpha_i \in (0, 1)$ and $n \geq 3$. Then there exists a unique Kähler metric on $X = \mathbb{CP}^1$ with $K_g = 1$ and cone angle $2\pi\alpha_i$ at x_i if and only if

$$\forall i \quad a_i < \frac{1}{2} \cdot \sum_{i=1}^n a_i \quad \text{where } a_i = 1 - \alpha_i$$

Main result

Main result

Theorem (Panov 2009 when $d = 2$, dB-Panov 2025-26 for $d \geq 2$)

Let $\mathcal{H} = \{H_1, \dots, H_n\}$ be a hyperplane arrangement in $\mathbb{CP}^d$ with $d \geq 1$.

Main result

Theorem (Panov 2009 when $d = 2$, dB-Panov 2025-26 for $d \geq 2$)

Let $\mathcal{H} = \{H_1, \dots, H_n\}$ be a hyperplane arrangement in $\mathbb{CP}^d$ with $d \geq 1$. Let $\alpha_1, \dots, \alpha_n \in (0, 1)$. Write $a_i = 1 - \alpha_i$.

Main result

Theorem (Panov 2009 when $d = 2$, dB-Panov 2025-26 for $d \geq 2$)

Let $\mathcal{H} = \{H_1, \dots, H_n\}$ be a hyperplane arrangement in $\mathbb{CP}^d$ with $d \geq 1$. Let $\alpha_1, \dots, \alpha_n \in (0, 1)$. Write $a_i = 1 - \alpha_i$. Assume that:

Main result

Theorem (Panov 2009 when $d = 2$, dB-Panov 2025-26 for $d \geq 2$)

Let $\mathcal{H} = \{H_1, \dots, H_n\}$ be a hyperplane arrangement in $\mathbb{CP}^d$ with $d \geq 1$. Let $\alpha_1, \dots, \alpha_n \in (0, 1)$. Write $a_i = 1 - \alpha_i$. Assume that:

① $\sum_{i=1}^n a_i = d + 1$

Main result

Theorem (Panov 2009 when $d = 2$, dB-Panov 2025-26 for $d \geq 2$)

Let $\mathcal{H} = \{H_1, \dots, H_n\}$ be a hyperplane arrangement in $\mathbb{CP}^d$ with $d \geq 1$. Let $\alpha_1, \dots, \alpha_n \in (0, 1)$. Write $a_i = 1 - \alpha_i$. Assume that:

- ① $\sum_{i=1}^n a_i = d + 1$
- ② the weighted arrangement $\{(H_i, a_i)\}_{i=1}^n$ is **stable**

Main result

Theorem (Panov 2009 when $d = 2$, dB-Panov 2025-26 for $d \geq 2$)

Let $\mathcal{H} = \{H_1, \dots, H_n\}$ be a hyperplane arrangement in $\mathbb{CP}^d$ with $d \geq 1$. Let $\alpha_1, \dots, \alpha_n \in (0, 1)$. Write $a_i = 1 - \alpha_i$. Assume that:

- ① $\sum_{i=1}^n a_i = d + 1$
- ② the weighted arrangement $\{(H_i, a_i)\}_{i=1}^n$ is **stable**

Then the following inequality holds:

$$Q(a_1, \dots, a_n) \leq 0 \tag{1}$$

where Q is the **Hirzebruch quadratic form** of $\mathcal{H}$.

Main result

Theorem (Panov 2009 when $d = 2$, dB-Panov 2025-26 for $d \geq 2$)

Let $\mathcal{H} = \{H_1, \dots, H_n\}$ be a hyperplane arrangement in $\mathbb{CP}^d$ with $d \geq 1$. Let $\alpha_1, \dots, \alpha_n \in (0, 1)$. Write $a_i = 1 - \alpha_i$. Assume that:

- ① $\sum_{i=1}^n a_i = d + 1$
- ② the weighted arrangement $\{(H_i, a_i)\}_{i=1}^n$ is **stable**

Then the following inequality holds:

$$Q(a_1, \dots, a_n) \leq 0 \tag{1}$$

where Q is the **Hirzebruch quadratic form** of $\mathcal{H}$. Moreover, the equality holds if and only if there exists a **polyhedral Kähler (PK)** metric on $\mathbb{CP}^d$ with cone angles $2\pi\alpha_i$ along H_i .

Let σ_i be the number of points p with $r_p \geq 3$ lying on the i -th line of the given arrangement of k lines in the plane. We consider the $(k \times k)$ -symmetric matrix A with

$$(3) \quad A_{ij} = \begin{cases} 3\sigma_i - 4 & (i = j), \\ 2 & (i \neq j, \quad p \in L_i \cap L_j \text{ with } r_p = 2), \\ -1 & (i \neq j, \quad p \in L_i \cap L_j \text{ with } r_p \geq 3). \end{cases}$$

With the k lines we associate real variables x_i and let x be the column vector $(x_1, \dots, x_k)$. With the s points p_j with $r_{p_j} \geq 3$ we associate real variables y_j .

For each point p_j with $r_{p_j} \geq 3$ we consider the linear form

$$P_j(x, y) = 2y_j + \sum_{p_j \in L_i} x_i, \quad \text{where } y = (y_1, \dots, y_s).$$

Höfer's formula. For the algebraic surface Y (a good covering of S of degree d with respect to $L_1, \dots, L_k, E_1, \dots, E_s$ and the given branching numbers $n_1, \dots, n_k, m_1, \dots, m_s$) we have

$$(4) \quad (3c_2(Y) - c_1^2(Y))/d = \frac{1}{4} \left(x^t A x + \sum_{j=1}^s P_j(x, y)^2 \right),$$

where $x_i = 1 - \frac{1}{n_i}$ and $y_j = -1 - \frac{1}{m_j}$.

Sketch proof

Follow Panov (2009)

Sketch proof

Follow Panov (2009)

- Take a log resolution $\pi : X \rightarrow \mathbb{CP}^d$

Sketch proof

Follow Panov (2009)

- Take a log resolution $\pi : X \rightarrow \mathbb{CP}^d$
- Define a parabolic structure E_* on $E = \pi^*T\mathbb{CP}^d$

Sketch proof

Follow Panov (2009)

- Take a log resolution $\pi : X \rightarrow \mathbb{CP}^d$
- Define a parabolic structure E_* on $E = \pi^*T\mathbb{CP}^d$
 - ① $\text{parc}_1(E_*) = 0 \iff \sum_{i=1}^n a_i = d + 1$

Sketch proof

Follow Panov (2009)

- Take a log resolution $\pi : X \rightarrow \mathbb{CP}^d$
- Define a parabolic structure E_* on $E = \pi^*T\mathbb{CP}^d$
 - ① $\text{parc}_1(E_*) = 0 \iff \sum_{i=1}^n a_i = d + 1$
 - ② E_* is slope stable $\iff \{(H_i, a_i)\}_{i=1}^n$ is stable

Sketch proof

Follow Panov (2009)

- Take a log resolution $\pi : X \rightarrow \mathbb{CP}^d$
- Define a parabolic structure E_* on $E = \pi^*T\mathbb{CP}^d$
 - ① $\text{parc}_1(E_*) = 0 \iff \sum_{i=1}^n a_i = d + 1$
 - ② E_* is slope stable $\iff \{(H_i, a_i)\}_{i=1}^n$ is stable
 - ③ $\text{parch}_2(E_*) = 0 \iff Q(a_1, \dots, a_n) = 0$

Sketch proof

Follow Panov (2009)

- Take a log resolution $\pi : X \rightarrow \mathbb{CP}^d$
- Define a parabolic structure E_* on $E = \pi^*T\mathbb{CP}^d$
 - ① $\text{parc}_1(E_*) = 0 \iff \sum_{i=1}^n a_i = d + 1$
 - ② E_* is slope stable $\iff \{(H_i, a_i)\}_{i=1}^n$ is stable
 - ③ $\text{parch}_2(E_*) = 0 \iff Q(a_1, \dots, a_n) = 0$
- Apply the Kobayashi-Hitchin correspondence for parabolic bundles (Mochizuki, 2006) to obtain a *flat unitary logarithmic connection* $\tilde{\nabla}$ on E adapted to the parabolic structure

Sketch proof

Follow Panov (2009)

- Take a log resolution $\pi : X \rightarrow \mathbb{CP}^d$
- Define a parabolic structure E_* on $E = \pi^*T\mathbb{CP}^d$
 - ① $\text{parc}_1(E_*) = 0 \iff \sum_{i=1}^n a_i = d + 1$
 - ② E_* is slope stable $\iff \{(H_i, a_i)\}_{i=1}^n$ is stable
 - ③ $\text{parch}_2(E_*) = 0 \iff Q(a_1, \dots, a_n) = 0$
- Apply the Kobayashi-Hitchin correspondence for parabolic bundles (Mochizuki, 2006) to obtain a *flat unitary logarithmic connection* $\tilde{\nabla}$ on E adapted to the parabolic structure
- $\tilde{\nabla} = \pi^*\nabla$ where ∇ is a torsion-free (flat, unitary, logarithmic) connection on $T\mathbb{CP}^n$

Sketch proof

Follow Panov (2009)

- Take a log resolution $\pi : X \rightarrow \mathbb{CP}^d$
- Define a parabolic structure E_* on $E = \pi^*T\mathbb{CP}^d$
 - ① $\text{parc}_1(E_*) = 0 \iff \sum_{i=1}^n a_i = d + 1$
 - ② E_* is slope stable $\iff \{(H_i, a_i)\}_{i=1}^n$ is stable
 - ③ $\text{parch}_2(E_*) = 0 \iff Q(a_1, \dots, a_n) = 0$
- Apply the Kobayashi-Hitchin correspondence for parabolic bundles (Mochizuki, 2006) to obtain a *flat unitary logarithmic connection* $\tilde{\nabla}$ on E adapted to the parabolic structure
- $\tilde{\nabla} = \pi^*\nabla$ where ∇ is a torsion-free (flat, unitary, logarithmic) connection on $T\mathbb{CP}^n$
- ∇ is the Levi-Civita connection of a flat Kähler metric g on the arrangement complement whose metric completion is polyhedral

Questions

Questions

- Which arrangements admit PK metrics?

Questions

- Which arrangements admit PK metrics?
- Suppose (X, Δ) is a klt pair endowed with a weak KE metric g

Questions

- Which arrangements admit PK metrics?
- Suppose (X, Δ) is a klt pair endowed with a weak KE metric g
 - ① When does g have constant holomorphic sectional curvature?

Questions

- Which arrangements admit PK metrics?
- Suppose (X, Δ) is a klt pair endowed with a weak KE metric g
 - ① When does g have constant holomorphic sectional curvature?
 - ② Is $|\text{Riem}(g)|^2$ locally integrable near the conical divisor?

Questions

- Which arrangements admit PK metrics?
- Suppose (X, Δ) is a klt pair endowed with a weak KE metric g
 - ① When does g have constant holomorphic sectional curvature?
 - ② Is $|\text{Riem}(g)|^2$ locally integrable near the conical divisor?
 - ③ If g is flat, does it need to be polyhedral?

Questions

- Which arrangements admit PK metrics?
- Suppose (X, Δ) is a klt pair endowed with a weak KE metric g
 - ① When does g have constant holomorphic sectional curvature?
 - ② Is $|\text{Riem}(g)|^2$ locally integrable near the conical divisor?
 - ③ If g is flat, does it need to be polyhedral?
- Are there any other examples of PK manifolds with cone angles less than 2π besides complex projective space and products?

- Which arrangements admit PK metrics?
- Suppose (X, Δ) is a klt pair endowed with a weak KE metric g
 - ① When does g have constant holomorphic sectional curvature?
 - ② Is $|\text{Riem}(g)|^2$ locally integrable near the conical divisor?
 - ③ If g is flat, does it need to be polyhedral?
- Are there any other examples of PK manifolds with cone angles less than 2π besides complex projective space and products?

THANK YOU!